Proof

The multiplicative inverse is less obvious than the additive one.

Let $z^{-1} = u + iv$ be the inverse of $z = x + iy$

$$\text{We have} \quad z\,z^{-1} = 1$$

$$\text{That is} \quad (x+iy)(u+iv) = 1$$

$$(xu - yv) + i(xv + uy) = 1 + i0$$

Equating real and imaginary parts we get

$$xu - yv = 1 \text{ and } xv + uy = 0\,.$$

Solving the above system of simultaneous equations in u and v

we get $u = \dfrac{x}{x^2+y^2}$ and $v = \dfrac{-y}{x^2+y^2}$. $(\because z \text{ is non-zero} \Rightarrow x^2 + y^2 > 0)$

If $z = x + iy$, then $z^{-1} = \dfrac{x}{x^2+y^2} + i\dfrac{-y}{x^2+y^2}$. $(\because z^{-1} \text{ is not defined when } z = 0)$. ■

Note that the above example shows the existence of z^{-1} of the complex number z. To compute the inverse of a given complex number, we conveniently use $z^{-1} = \dfrac{1}{z}$. If z_1 and z_2 are two complex numbers where $z_2 \neq 0$, then the product of z_1 and $\dfrac{1}{z_2}$ is denoted by $\dfrac{z_1}{z_2}$. Other properties can be verified in a similar manner. In the next section, we define the conjugate of a complex number. It would help us to find the inverse of a complex number easily.

Complex numbers obey the laws of indices

(i) $z^m z^n = z^{m+n}$ (ii) $\dfrac{z^m}{z^n} = z^{m-n}$, $z \neq 0$ (iii) $\left(z^m\right)^n = z^{mn}$ (iv) $\left(z_1 z_2\right)^m = {z_1}^m {z_2}^m$

EXERCISE 2.3

1. If $z_1 = 1 - 3i$, $z_2 = -4i$, and $z_3 = 5$, show that

   (i) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ (ii) $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.

2. If $z_1 = 3$, $z_2 = -7i$, and $z_3 = 5 + 4i$, show that

   (i) $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ (ii) $(z_1 + z_2) z_3 = z_1 z_3 + z_2 z_3$.

3. If $z_1 = 2 + 5i$, $z_2 = -3 - 4i$, and $z_3 = 1 + i$, find the additive and multiplicative inverse of z_1, z_2, and z_3.

2.4 Conjugate of a Complex Number

In this section, we study about conjugate of a complex number, its geometric representation, and properties with suitable examples.

Definition 2.3

The conjugate of the complex number $x + iy$ is defined as the complex number $x - iy$.

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